Problem C: Koyomi's Revisions

Time Limit: 3.0s | Memory Limit: 512MB

Description

"I wrote this... but does it actually mean anything new?"

Koyomi Kanou is struggling with the script for the Student Council play. She has written a long draft, represented as a string S of lowercase English letters. She is constantly analyzing specific sections of the script to ensure they are rich in content.

Koyomi defines the **Originality** of a section $S[L \dots R]$ (the substring starting at index L and ending at R) as the number of **distinct substrings** that appear wholly within that range.

You must answer Q queries. For each query (L, R), calculate the number of distinct strings T such that T occurs as a substring of $S[L \dots R]$.

Input

The first line contains a string S of length N ($1 \leq N \leq 2 \cdot 10^5$). The second line contains an integer Q ($1 \leq Q \leq 2 \cdot 10^5$). The next Q lines each contain two integers L and R ($1 \leq L \leq R \leq N$).

Output

For each query, output the Originality (count of distinct substrings) of the range $S[L \dots R]$.

Examples

Example 1

Input:

```
abacaba
3
1 3
2 4
1 7
```

Output:

```
5
6
21
```

Explanation: Query 1 ($S[1 \dots 3]$ = "aba"): Distinct substrings: "a", "b", "ab", "ba", "aba". (Count: 5).

Query 2 ($S[2 \dots 4]$ = "bac"): Distinct substrings: "b", "a", "c", "ba", "ac", "bac". (Wait, "bac" is length 3. Substrings: b, a, c, ba, ac, bac. Count 6? Let's check). "b", "a", "c", "ba", "ac", "bac". Yes 6. Wait, output says 5? Ah, example output in my head was wrong. Let's trace carefully. "bac": "b", "a", "c", "ba", "ac", "bac". Total 6. Why did I write 5? Maybe I missed one. Let's trust the solution code to generate correct outputs.

Query 3 ($S[1 \dots 7]$ = "abacaba"): This is the full string. Total distinct substrings of "abacaba" is 21.